Project GyerAR: A Low-Cost Augmented Reality Smart Glasses Architecture Built on First Principles

DIY Engineering White Paper

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Abstract

This document evaluates the theoretical and practical feasibility of an ultra-low-cost, functional augmented reality (AR) and camera-enabled smart glasses platform. Throughout the development phase, the *First Principles* methodology is applied to deconstruct complex, modern commercial AR systems into their bare minimum, core physical and software building blocks. The finalized architecture yields an estimated hardware bill of materials (BOM) cost ranging between 14,000 and 18,000 HUF (approx. 40 – 50 USD).

1 Introduction and First Principles

Modern commercial AR headsets (e.g., Apple Vision Pro, Microsoft HoloLens) rely on intricate waveguide optics, spatial computing chipsets, and dedicated depth sensors. By deconstructing the problem down to its fundamental physical requirements, a functional system must satisfy only the following four core constraints:

1. **Image Acquisition:** An optical sensor to digitalize the surrounding environment.
2. **Compute Unit:** A microcontroller to handle network communication and manage peripherals.
3. **Display System:** An emissive light source that projects information into the retina without obstructing ambient light from the real world.
4. **Power Supply:** A portable, lightweight electrochemical cell.

2 Hardware Architecture and Budget

The minimum viable components required to achieve functionality, along with their estimated market prices, are structured in Table 1.

3 Optical and Physical Implementation

3.1 The Pepper's Ghost Optical Alignment

Heads-up transparency is achieved utilizing the classic *Pepper's Ghost* illusion. The OLED display is mounted facing downwards on the upper rim of the glasses frame (near the brow line), projecting light vertically. A semi-reflective, transparent polymer sheet (e.g., thin acrylic) is positioned directly underneath at a 45° angle relative to the user's line of sight.

Based on the law of reflection (where the angle of incidence equals the angle of reflection), the display image undergoes a 90° directional shift straight into the retina, while the photons arriving from the background pass through the polymer matrix unhindered.

Table 1: Minimum Hardware Bill of Materials (BOM)

Component	**Type / Description**	**Estimated Price (HUF)**
Compute Unit + Camera	ESP32-CAM module (with OV2640 sensor)	2,500 – 3,500
Display	0.96" or 1.3" I2C monochrome OLED	1,500 – 2,500
Battery	Li-Po cell (150–300 mAh) + TP4056 charger	1,500 – 2,500
Structural Frame	Safety glasses or clear fashion glasses	1,000 – 2,000
Optical Combiner	Transparent rigid polymer sheet (CD case)	0 – 1,000
Consumables	Wires, adhesives, electrical tape	1,000
Total		**14,000 – 18,000**

3.2 The Focal Accommodation Dilemma

Because the polymer combiner sits mere centimeters away from the eye, the human eye cannot simultaneously accommodate (focus) on the far-field environment and the near-field virtual image. This mechanical trade-off is mitigated by extending the display assembly further forward along the temple arm of the frame, thereby minimizing ciliary muscle strain.

4 Software Architecture and Data Pipeline

The software implementation relies entirely on open-source and free libraries. The data pipeline is bifurcated into two primary execution layers:

1. **Firmware (C++/Arduino):** The ESP32-CAM operates as a local Wi-Fi Access Point. It broadcasts an MJPEG video stream via HTTP protocol to the receiving host unit.

2. **Processing and AI Layer (Python/OpenCV):** An external computer or smartphone ingests the network stream, executes computer vision workloads (e.g., via a YOLOv8 object detection pipeline), and transmits the processed metadata (such as object labels or distance estimates) back to the ESP32 via Bluetooth or UDP packets to drive the OLED display.

Since the physical combiner optics produce a mirrored reflection, the software must explicitly execute a horizontal coordinate inversion matrix-transformation during the OLED rendering pass:

$$I_{\text{render}}(x, y) = I_{\text{buffer}}(W - 1 - x, y) \tag{1}$$

where W represents the display width in pixels.

5 Conclusion

Project GyerAR demonstrates that the fundamental features of augmented reality (transparent displays and computer vision analytics) can be achieved without high-tech, proprietary hardware components. Although ergonomics, battery runtime (30–60 minutes), and industrial aesthetics face heavy compromises, the system delivers exceptional educational and hacker value.